PRAKTIKUM SISTEM OPERASI

MODUL 5

EKSPLORASI PROSES XINU

Oleh:

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PROGRAM STUDI S1 REKAYASA PERANGKAT LUNAK

DIREKTORAT KAMPUS PURWOKERTO

UNIVERSITAS TELKOM

2026

JURNAL

1. **[10 Poin]** Jawablah pertanyaan berikut ini:
    a. Berapa banyaknya maksimum proses yang ada pada Xinu?
    Jawaban: Maksimum jumlah proses pada Xinu adalah 8 proses, yang didefinisikan dengan konstanta NPROC = 8

    b. Berapa maksimal panjang nama suatu proses pada Xinu?
    Jawaban: 16 karakter define PNMLEN 16 termasuk karakter akhir (\0)

    c. Berapa nilai prioritas awal pada saat proses dibuat?
    Jawaban: 20 (Fungsi create() secara default menetapkan prioritas PR_DEFAULT = 20)

    d. Ada berapa total state pada Xinu? Sebutkan!
    Jawaban: 7 state
    Yaitu:
        1. PR_FREE: proses kosong / tidak digunakan
        2. PR_CURR: sedang berjalan (running)
        3. PR_READY: siap dijalankan
        4. PR_RECV: menunggu pesan
        5. PR_SLEEP: sedang tidur
        6. PR_SUSP: ditangguhkan
        7. PR_WAIT: menunggu semaphore

2. **[20 Poin]** Perintah ps adalah perintah untuk menampilkan statistik process yang berjalan. Source code dari ps tersimpan pada file xsh_ps.c. Carilah file tersebut dan beri komentar pada 20 baris terakhir di source code tersebut!
    Jawaban:

Activities Terminal Sat 11 Apr, 08:04

xinu@xinu-develop-end: ~/xinu

File Edit View Search Terminal Help

```
xinu@xinu-develop-end:~$ find / -name "xsh_.c" 2>/dev/null
xinu@xinu-develop-end:~$ ls
prev-version  README  xinu
xinu@xinu-develop-end:~$ cd xinu
xinu@xinu-develop-end:~/xinu$ ls
banner.h  config     DESCRIPTION  include  main.c  shell
compile   COPYRIGHT  device       lib      net     system
xinu@xinu-develop-end:~/xinu$ find . -name "xsh_ps.c"
./shell/xsh_ps.c
xinu@xinu-develop-end:~/xinu$ cat -n ./shell/xsh_ps.c
     1  /* xsh_ps.c - xsh_ps */
     2
     3  #include <xinu.h>
     4  #include <stdio.h>
     5  #include <string.h>
     6
     7  /*------------------------------------------------------------------------
     8   * xsh_ps - shell command to print the process table
     9   *------------------------------------------------------------------------
    10   */
    11  shellcmd xsh_ps(int nargs, char *args[])
    12  {
    13          struct  procent *prptr;         /* Pointer to process           */
    14          int32   i;                      /* Index into proctabl          */
    15          char *pstate[]  = {             /* Names for process states     */
    16                  "free ", "curr ", "ready", "recv ", "sleep", "susp ",
    17                  "wait ", "rtime"};
    18
```

Activities Terminal Sat 11 Apr, 08:04

xinu@xinu-develop-end: ~/xinu

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```
    19          /* For argument '--help', emit help about the 'ps' command      */
    20
    21          if (nargs == 2 && strncmp(args[1], "--help", 7) == 0) {
    22                  printf("Use: %s\n\n", args[0]);
    23                  printf("Description:\n");
    24                  printf("\tDisplays information about running processes\n");
    25                  printf("Options:\n");
    26                  printf("\t--help\t display this help and exit\n");
    27                  return 0;
    28          }
    29
    30          /* Check for valid number of arguments */
    31
    32          if (nargs > 1) {
    33                  fprintf(stderr, "%s: too many arguments\n", args[0]);
    34                  fprintf(stderr, "Try '%s --help' for more information\n",
    35                                  args[0]);
    36                  return 1;
    37          }
    38
    39          /* Print header for items from the process table */
    40
    41          printf("%3s %-16s %5s %4s %4s %10s %-10s %10s\n",
    42                     "Pid", "Name", "State", "Prio", "Ppid", "Stack Base",
    43                     "Stack Ptr", "Stack Size");
    44
    45          printf("%3s %-16s %5s %4s %4s %10s %-10s %10s\n",
    46                     "---", "----------------", "-----", "----", "----",
```

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xinu@xinu-develop-end: ~/xinu

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```
                return 1;
        }

        /* Print header for items from the process table */

        printf("%3s %-16s %5s %4s %4s %10s %-10s %10s\n",
                   "Pid", "Name", "State", "Prio", "Ppid", "Stack Base",
                   "Stack Ptr", "Stack Size");

        printf("%3s %-16s %5s %4s %4s %10s %-10s %10s\n",
                   "---", "----------------", "-----", "----", "----",
                   "----------", "----------", "----------");

        /* Output information for each process */

        for (i = 0; i < NPROC; i++) {
                prptr = &proctab[i];
                if (prptr->prstate == PR_FREE) {  /* Skip unused slots  */
                        continue;
                }
                printf("%3d %-16s %s %4d %4d 0x%08X 0x%08X %8d\n",
                        i, prptr->prname, pstate[(int)prptr->prstate],
                        prptr->prprio, prptr->prparent, prptr->prstkbase,
                        prptr->prstkptr, prptr->prstklen);
        }
        return 0;
}
xinu@xinu-develop-end:~/xinu$
```

Activities Terminal Sat 11 Apr, 08:31

xinu@xinu-develop-end: ~/xinu

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GNU nano 2.2.6 File: ./shell/xsh_ps.c Modified

```
        }

        /* Print header for items from the process table */

        printf("%3s %-16s %5s %4s %4s %10s %-10s %10s\n",
                   "Pid", "Name", "State", "Prio", "Ppid", "Stack Base",
                   "Stack Ptr", "Stack Size");

        printf("%3s %-16s %5s %4s %4s %10s %-10s %10s\n",
                   "---", "----------------", "-----", "----", "----",
                   "----------", "----------", "----------");

        /* Output information for each process */

        for (i = 0; i < NPROC; i++) {
                prptr = &proctab[i];
                if (prptr->prstate == PR_FREE) {  /* Skip unused slots  */
                        continue;
                }
                printf("%3d %-16s %s %4d %4d 0x%08X 0x%08X %8d\n",
                        i, prptr->prname, pstate[(int)prptr->prstate],
                        prptr->prprio, prptr->prparent, prptr->prstkbase,
                        prptr->prstkptr, prptr->prstklen);
```

^G Get Help ^O WriteOut ^R Read File ^Y Prev Page ^K Cut Text ^C Cur Pos
^X Exit ^J Justify ^W Where Is ^V Next Page ^U UnCut Text ^T To Spell

```
Activities  Terminal  Sat 11 Apr, 08:35
xinu@xinu-develop-end: ~/xinu
File  Edit  View  Search  Terminal  Help
  GNU nano 2.2.6            File: ./shell/xsh_ps.c                    Modified

        }

        /* Print header for items from the process table */
        /* Menampilkan header/judul kolom untuk tabel proses*/

        printf("%3s %-16s %5s %4s %4s %10s %-10s %10s\n",
                   "Pid", "Name", "State", "Prio", "Ppid", "Stack Base",
                   "Stack Ptr", "Stack Size");

        printf("%3s %-16s %5s %4s %4s %10s %-10s %10s\n",
                   "---", "----------------", "-----", "----", "----",
                   "----------", "----------", "----------");

        /* Output information for each process */
        /*Aulia Jasifa Br Ginting - 2311104060 - Jurnal 5*/

        for (i = 0; i < NPROC; i++) {
                prptr = &proctab[i];
                if (prptr->prstate == PR_FREE) {  /* Skip unused slots  */
                        continue;
                }
                printf("%3d %-16s %s %4d %4d 0x%08X 0x%08X %8d\n",
                        i, prptr->prname, pstate[(int)prptr->prstate],

^G Get Help   ^O WriteOut   ^R Read File  ^Y Prev Page  ^K Cut Text   ^C Cur Pos
^X Exit       ^J Justify    ^W Where Is   ^V Next Page  ^U UnCut Text ^T To Spell
```

```
Activities  Terminal  Sat 11 Apr, 08:45
xinu@xinu-develop-end: ~/xinu
File  Edit  View  Search  Terminal  Help
        /* Print header for items from the process table */
        /* Menampilkan header/judul kolom untuk tabel proses*/

        printf("%3s %-16s %5s %4s %4s %10s %-10s %10s\n",
                   "Pid", "Name", "State", "Prio", "Ppid", "Stack Base",
                   "Stack Ptr", "Stack Size");

        printf("%3s %-16s %5s %4s %4s %10s %-10s %10s\n",
                   "---", "----------------", "-----", "----", "----",
                   "----------", "----------", "----------");

        /* Output information for each process */
        /*Aulia Jasifa Br Ginting - 2311104060 - Jurnal 5*/

        for (i = 0; i < NPROC; i++) {
                prptr = &proctab[i];
                if (prptr->prstate == PR_FREE) {  /* Skip unused slots  */
                        continue;
                }
                printf("%3d %-16s %s %4d %4d 0x%08X 0x%08X %8d\n",
                        i, prptr->prname, pstate[(int)prptr->prstate],
                        prptr->prprio, prptr->prparent, prptr->prstkbase,
                        prptr->prstkptr, prptr->prstklen);
        }
        return 0;

}
xinu@xinu-develop-end:~/xinu$
```

3. **[35 Poin]** Ubahlah perintah ps (source code: xsh_ps.c) pada Xinu sehingga menampilkan informasi tambahan berupa kolom yang berisi total message yang ada pada proses seperti gambar di bawah ini:

```
xsh $ ps
Pid Name             State Prio Ppid Stack Base Stack Ptr  Stack Size Msg Content
--- ---------------- ----- ---- ---- ---------- ---------- ---------- --- -------
  0 prnull           ready    0    0 0x005FDFFC 0x00FFFF04     8192   1       4
  1 rdsproc          susp   200    0 0x005FBFFC 0x005FBFC8    16384   0       0
  2 ipout            wait   500    0 0x005F7FFC 0x005F7F40     8192   0       0
  3 netin            wait   500    0 0x005F5FFC 0x005F5EE0     8192   0       0
  5 Main process     recv    20    4 0x005E3FFC 0x005E3F64    65536   0       0
  6 shell            recv    50    5 0x005D3FFC 0x005D3C7C     8192   0       0
  7 ps               curr    20    6 0x005F3FFC 0x005F3DC8     8192   0       0
```

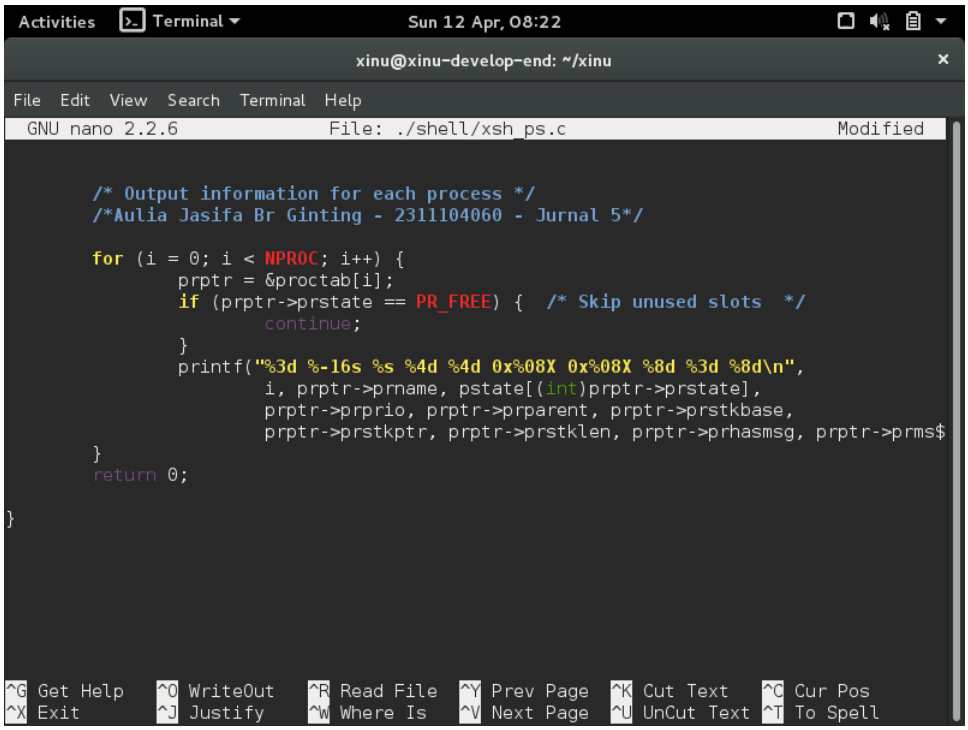
Kolom Msg adalah banyaknya pesan yang ada dalam proses.

Kolom Content adalah isi dari pesan tersebut.

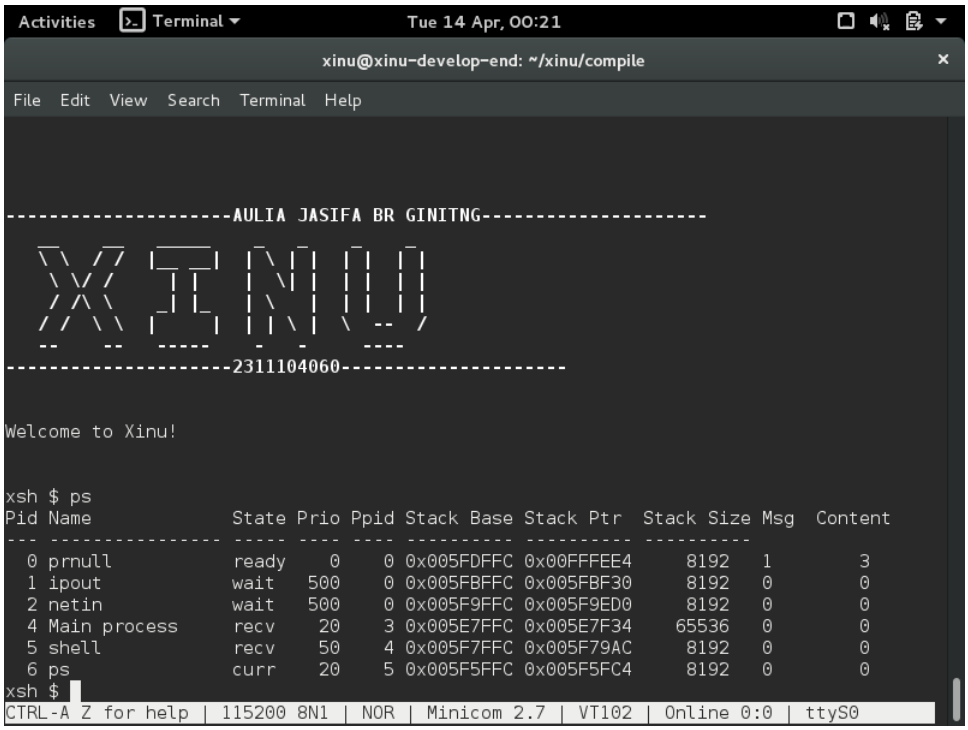
Langkah pengerjaan:

- Modifikasi source code pada file xsh_ps.c

```
Activities  Terminal   Sun 12 Apr, 08:22
xinu@xinu-develop-end: ~/xinu
File  Edit  View  Search  Terminal  Help
GNU nano 2.2.6          File: ./shell/xsh_ps.c                    Modified

        }

        /* Print header for items from the process table */
        /* Menampilkan header/judul kolom untuk tabel proses*/

        printf("%3s %-16s %5s %4s %4s %10s %-10s %10s %3s %8s\n",
                   "Pid", "Name", "State", "Prio", "Ppid", "Stack Base",
                   "Stack Ptr", "Stack Size", "Msg", "Content");

        printf("%3s %-16s %5s %4s %4s %10s %-10s %10s\n",
                   "---", "----------------", "-----", "----", "----",
                   "----------", "----------", "----------", "---", "--------");

        /* Output information for each process */
        /*Aulia Jasifa Br Ginting - 2311104060 - Jurnal 5*/

        for (i = 0; i < NPROC; i++) {
                prptr = &proctab[i];
                if (prptr->prstate == PR_FREE) {  /* Skip unused slots  */
                        continue;
                }
                printf("%3d %-16s %s %4d %4d 0x%08X 0x%08X %8d %3d %8d\n",
                        i, prptr->prname, pstate[(int)prptr->prstate],

^G Get Help   ^O WriteOut   ^R Read File   ^Y Prev Page   ^K Cut Text   ^C Cur Pos
^X Exit       ^J Justify    ^W Where Is    ^V Next Page   ^U UnCut Text ^T To Spell
```

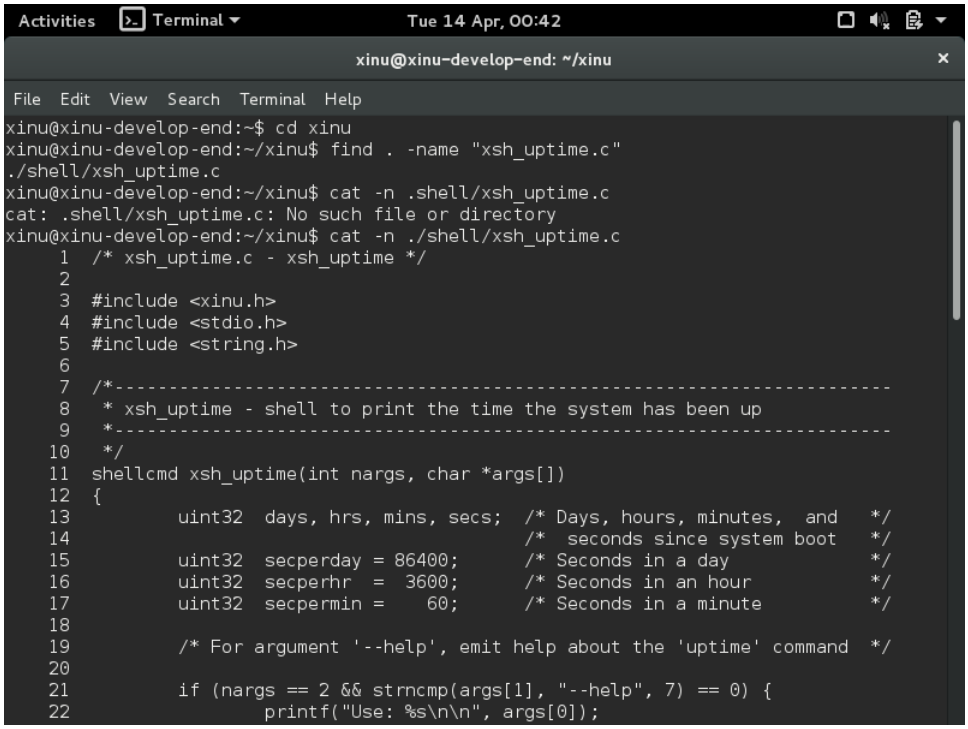


- Kompilasi ulang Xinu dengan perintah seperti pada modul sebelumnya
- Jalankan Backend VM
- Setelah sistem berjalan, jalankan perintah $ps. Pastikan hasilnya sesuai dengan contoh output pada gambar yang diberikan.



- Screenshot source kode dan output akhir hasil modifikasi

4. **[35 Poin]** Ubahlah perintah uptime pada Xinu sehingga menampilkan lamanya Xinu sejak booting hanya dalam satuan menit.
   Langkah pengerjaan:
   - Modifikasi source code pada file xsh_uptime.c
   - Kompilasi ulang Xinu dengan perintah seperti pada modul sebelumnya
   - Jalankan Backend VM
   - Setelah sistem berjalan, jalankan perintah $uptime. Pastikan hasilnya sesuai dengan contoh output yang diinginkan
   - Screenshot source kode dan output akhir hasil modifikasi



Activities Terminal Tue 14 Apr, 00:48

xinu@xinu-develop-end: ~/xinu

File Edit View Search Terminal Help

GNU nano 2.2.6 File: ./shell/xsh_uptime.c Modified

```c
/* xsh_uptime.c - xsh_uptime */

#include <xinu.h>
#include <stdio.h>
#include <string.h>

/*------------------------------------------------------------------------
 * xsh_uptime - shell to print the time the system has been up
 *------------------------------------------------------------------------
 */
shellcmd xsh_uptime(int nargs, char *args[])
{
        uint32  days, hrs, mins, secs;  /* Days, hours, minutes,  and   */
                                        /*  seconds since system boot   */
        uint32  secperday = 86400;      /* Seconds in a day             */
        uint32  secperhr  =  3600;      /* Seconds in an hour           */
        uint32  secpermin =    60;      /* Seconds in a minute          */

        /* For argument '--help', emit help about the 'uptime' command  */

        if (nargs == 2 && strncmp(args[1], "--help", 7) == 0) {
                printf("Use: %s\n\n", args[0]);
                printf("Description:\n");
```

^G Get Help ^O WriteOut ^R Read File ^Y Prev Page ^K Cut Text ^C Cur Pos
^X Exit ^J Justify ^W Where Is ^V Next Page ^U UnCut Text ^T To Spell

Activities Terminal Tue 14 Apr, 00:47

xinu@xinu-develop-end: ~/xinu

File Edit View Search Terminal Help

GNU nano 2.2.6 File: ./shell/xsh_uptime.c Modified

```c
/* xsh_uptime.c - xsh_uptime */

#include <xinu.h>
#include <stdio.h>
#include <string.h>

/*------------------------------------------------------------------------
 * xsh_uptime - shell to print the time the system has been up
 *------------------------------------------------------------------------
 */

/* Aulia Jasifa Br Ginting - 2311104060 - Jurnal 5*/
shellcmd xsh_uptime(int nargs, char *args[])
{
        uint32  days, hrs, mins, secs;  /* Days, hours, minutes,  and   */
                                        /*  seconds since system boot   */
        uint32  secperday = 86400;      /* Seconds in a day             */
        uint32  secperhr  =  3600;      /* Seconds in an hour           */
        uint32  secpermin =    60;      /* Seconds in a minute          */

        /* For argument '--help', emit help about the 'uptime' command  */

        if (nargs == 2 && strncmp(args[1], "--help", 7) == 0) {
```

^G Get Help ^O WriteOut ^R Read File ^Y Prev Page ^K Cut Text ^C Cur Pos
^X Exit ^J Justify ^W Where Is ^V Next Page ^U UnCut Text ^T To Spell

```
Activities  Terminal  Tue 14 Apr, 00:51
xinu@xinu-develop-end: ~/xinu/compile
File  Edit  View  Search  Terminal  Help
    74          return 0;
    75  }
xinu@xinu-develop-end:~/xinu$ nano ./shell/xsh_uptime.c
xinu@xinu-develop-end:~/xinu$ cd compile
xinu@xinu-develop-end:~/xinu/compile$ make clean
removing .o files
removing configuration files ...
make[1]: Entering directory '/home/xinu/xinu/config'
rm -f config lex.yy.c y.tab.c
make[1]: Leaving directory '/home/xinu/xinu/config'
removing xinu ...
xinu@xinu-develop-end:~/xinu/compile$ make
making the config program
make[1]: Entering directory '/home/xinu/xinu/config'
rm -f config lex.yy.c y.tab.c
flex config.l
bison -y         config.y
gcc -o config y.tab.c  -lfl
./config Configuration conf.c conf.h
cp -p conf.h ../include
cp -p conf.c ../system
make[1]: Leaving directory '/home/xinu/xinu/config'
making ../system/conf.c
make[1]: Entering directory '/home/xinu/xinu/compile'
forcing a rebuild of conf.h and conf.c
make[2]: Entering directory '/home/xinu/xinu/compile'
creating new version
make[2]: Leaving directory '/home/xinu/xinu/compile'
```

```
Activities  Terminal  Tue 14 Apr, 00:51
xinu@xinu-develop-end: ~/xinu/compile
File  Edit  View  Search  Terminal  Help
          [0x00100000 to 0x005FBFFF]
    115657 bytes of Xinu code.
          [0x00100000 to 0x0011C3C8]
    148488 bytes of data.
          [0x0011FEC0 to 0x001442C7]

Obtained IP address  10.0.3.15   (0x0a00030f)




--------------------AULIA JASIFA BR GINITNG---------------------
   __    __   _____    _   _    _    _
   \ \   / /  |__ __|  | \ | |  | |  | |
    \ \/ /     | |     |  \| |  | |  | |
    / /\ \    _| |_    | \   |  | |  | |
   / /  \ \  |       |  | | \ |  \  --  /
   --    --   -----     -   -     ----
--------------------2311104060---------------------


Welcome to Xinu!


xsh $ uptime
Xinu has been up  1 second(s)
xsh $
CTRL-A Z for help | 115200 8N1 | NOR | Minicom 2.7 | VT102 | Offline | ttyS0
```

Activities Terminal Tue 14 Apr, 00:56

xinu@xinu-develop-end: ~/xinu

File Edit View Search Terminal Help

```
xinu@xinu-develop-end:~/xinu$ cat -n ./shell/xsh_uptime.c
     1  /* xsh_uptime.c - xsh_uptime */
     2
     3  #include <xinu.h>
     4  #include <stdio.h>
     5  #include <string.h>
     6
     7  /*------------------------------------------------------------------------
     8   * xsh_uptime - shell to print the time the system has been up
     9   *------------------------------------------------------------------------
    10   */
    11  /* Aulia Jasifa Br Ginting - 2311104060 - Jurnal 5 */
    12  shellcmd xsh_uptime(int nargs, char *args[])
    13  {
    14          uint32  days, hrs, mins, secs;  /* Days, hours, minutes,  and   */
    15                                          /*  seconds since system boot   */
    16          uint32  secperday = 86400;      /* Seconds in a day             */
    17          uint32  secperhr  =  3600;      /* Seconds in an hour           */
    18          uint32  secpermin =    60;      /* Seconds in a minute          */
    19
    20          /* For argument '--help', emit help about the 'uptime' command  */
    21
    22          if (nargs == 2 && strncmp(args[1], "--help", 7) == 0) {
    23                  printf("Use: %s\n\n", args[0]);
    24                  printf("Description:\n");
    25                  printf("\tDisplays time since the system booted\n");
    26                  printf("Options:\n");
    27                  printf("\t--help\t display this help and exit\n");
```